

# SecureMailScope

## SIH26159 — AI-Assisted Cryptographic Security Posture Assessment

Scan: SMS-PCAP-20260930-153242-ace8 | Mode: REAL\_PCAP\_SCAPY  
Generated: 2026-09-30T10:02:42.764710+00:00

Overall posture: 100/100 — GOOD

Sessions: 4 | Secure/Encrypted sessions: 4 | Findings: 0 | Critical: 0 | High: 0 | Medium: 0 | Low: 0

### Session Inventory

| Session | Protocol | TLS     | Cipher                                | Certificate | Packets |
|---------|----------|---------|---------------------------------------|-------------|---------|
| imap-1  | IMAP     | TLS 1.2 | TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 | COMBUSTION  | 6-10    |
| pop3-2  | POP3     | TLS 1.2 | TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 | COMBUSTION  | 11-15   |
| smtp-0  | SMTP     | TLS 1.2 | TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 | COMBUSTION  | 1-5     |
| smtp-3  | SMTP     | TLS 1.2 | TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 | COMBUSTION  | 16-20   |

### Findings

| ID | Severity | Session | Finding | Evidence |
|----|----------|---------|---------|----------|
|----|----------|---------|---------|----------|

## AI-Assisted Insights

- Isolation Forest highlights imap-1 as the highest-anomaly session with score 0.5.
- Risk conclusions are derived from observed protocol/cryptographic evidence; anomaly scores identify unusual profiles and do not by themselves prove compromise.

## Recommendations

### Integrity Evidence

Final evidence hash: 62035f9ba0fa9794f4c0549bb15c77f98cb3243efdbc0cbe199ef238d73afbea

Final audit record hash: 584f41925520cf77a2597bdcdaef17e3e9ec2f16085bea47b6c92affa77b6ec5

The current prototype uses a chained SHA-256 audit ledger. A production deployment can anchor the final evidence hash to a permissioned ledger.